Mth101 Midterm Paper 6 (Virtualustaad

Calculus (Virtual University of Pakistan)

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Solve by gulshan

MIDTERM EXAMINATION
Spring 2009
MTH101- Calculus And Analytical Geometry (Session - 6)
Time: 60 min

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Question No: 1 (Marks: 1) - Please choose one

The set {…,-4,-3,-2,-1,0,1,2,3,4,..} is know as set of ……..

- ► Natural numbers
- ► Integers
- ► Whole numbers
- ► None of these

Question No: 2 (Marks: 1) - Please choose one

The domain of the function $h(x) = \frac{1}{(x-2)(x-4)}$ is

- ► $(-\infty, 2) \cup (2,4) \cup (4, +\infty)$
- ► $(-\infty, 2\} \cup \{2,4\} \cup \{4, \infty)$
- ► $(-\infty, 2.5) \cup (2.5, 4.5) \cup (4.5, \infty)$
- ► All of these are incorrect

Question No: 3 (Marks: 1) - Please choose one

If the $\lim_{x \to a} f(x) = L$ then the inequality $(L-\varepsilon) < f(x) < L+\varepsilon$ holds in any subset of the interval

- ► $(a-\delta, a) \cup (a, a+\delta)$
- ► $(a-1, a) \cup (a, a+1)$

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► $(a - \varepsilon, a) \cup (a, a + \varepsilon)$

► None of these

Question No: 4 (Marks: 1) - Please choose one

$L - \varepsilon < f(x) < L + \varepsilon$ Can be written as

► $|f(x) - L| < \varepsilon$

► $|f(x) - L| > \varepsilon$

► $|f(x) - L| < \varepsilon + 1$

► None of these

Question No: 5 (Marks: 1) - Please choose one

If a function satisfies the conditions

f(c) is defined

Exists $\lim_{x \to c^+} f(x)$

$\lim_{x \to c^+} f(x) = f(c)$

Then the function is said to be

► Continuous at c

► Continuous from left at c

► Continuous from right at c

► None of these

Question No: 6 (Marks: 1) - Please choose one

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$\frac{d}{dx}[sex] =$- - - - -

- ▶ $\frac{\sin x}{1 - \sin^2 x}$
- ▶ $\frac{-\sin x}{1 - \sin^2 x}$
- ▶ $\frac{1}{1 - \sin^2 x}$
- ▶ None of these

Question No: 7 (Marks: 1) - Please choose one

$\log_b ac =$- - - - - -

- ▶ $\log_b a + \log_b c$
- ▶ $\log_a b + \log_c b$
- ▶ $\log_{a+c} b$
- ▶ None of these

Question No: 8 (Marks: 1) - Please choose one

$\log_b a^r =$- - - - - -

- ▶ $a \log_b r$
- ▶ $r \log_b a$
- ▶ $b \log_a r$
- ▶ None of these

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Question No: 9 (Marks: 1) - Please choose one

If $f''(x) < 0$ on an open interval (a,b) then f is ---------- on (a,b)

- ► None of these
- ► Concave up
- ► Concave down
- ► Closed

Question No: 10 (Marks: 1) - Please choose one

If f is a twice differentiable function at a stationary point x_0 and $f''(x_0) > 0$ then f has relative ..... At x_0

- ► Minima
- ► Maxima
- ► None of these

Question No: 11 (Marks: 1) - Please choose one

A line is called a tangent line to the circle if it meets the circle at precisely .....

- ► One point
- ► Two points
- ► Infinite points
- ► None of these

Question No: 12 (Marks: 1) - Please choose one

The equation $(x+4)^2 + (y-1)^2 = 6$ represents a circle having center at ..... and radius .....

- ► $(-4,1), \sqrt{6}$

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► $(-4,1),6$

► $(-4,-1),\sqrt{6}$

► None of these

Question No: 13 (Marks: 1) - Please choose one

The $\lim_{x\to a} f(x)\ where\ f(x)=k$ (k is a constant) is equal to

► k+2

► k+1

► k

► kf

Question No: 14 (Marks: 1) - Please choose one

For any polynomial $P(x)=c_0+c_1x+\ldots+c_nx^n$ and any real number a

$\lim_{x\to a} P(x)=c_0+c_1a+\ldots+c_na^n=$

► $P(a)$

► $P(a+1)$

► $P(a-1)$

► $\frac{1}{P(a)}$

Question No: 15 (Marks: 1) - Please choose one

Polynomials are always ..... Function

► Continuous

► Discontinuous

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Question No: 16 (Marks: 1) - Please choose one

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where d is a constant

$\frac{D}{Dx}[dh(x)] =$- - - -

- ► $dh(x)$
- ► $dh'(x)$
- ► 0
- ► None of these

Question No: 17 (Marks: 1) - Please choose one

The graph

$x = y^2$ is symmetric about

- ► X-axis
- ► Y-axis
- ► Origin
- ► None of these

Question No: 18 (Marks: 1) - Please choose one

Consider two function

$f(x) = 3\sqrt{x} and g(x) = \sqrt{x}$ what is true about these functions

- ► $f(x).g(x) = 3x$
- ► $f(x) / g(x) = 3x$
- ► $f(g(x)) = 3x$
- ► None of these

Question No: 19 (Marks: 1) - Please choose one

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The formula

$\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$ is called ..... with respect to x of the function f

- ► Derivative
- ► Slope
- ► Tangent
- ► None of these

Question No: 20 (Marks: 1) - Please choose one

Suppose that

$f \text{ and } g$ are differentiable function of x then $\frac{d}{dx}(\frac{f}{g})$

- ► $\frac{g.f'-f.g'}{g^2}$
- ► $\frac{g.f'+f.g'}{g^2}$
- ► $\frac{g.f'-f.g'}{g}$
- ► None of these

Question No: 21 (Marks: 2)

If $\frac{dy}{dx}=-\frac{3yx^2+1}{28y^3+x^3}$ then find the slope of the tangent line at the point (2, 0).

Question No: 22 (Marks: 3)

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$$Let \quad f(x) = \begin{cases} \dfrac{x^2 - x - 2}{x+1} & if \quad x \neq -1 \\ -3 & if \quad x = -1 \end{cases}$$

At what points the function f is continuous and discontinuous? At point of discontinuity if any explain why it is discontinuous?

Question No: 23 (Marks: 5)

Differentiate w.r.t. x by chain rule

$y = \sqrt{x^2 + 1}$

Question No: 24 (Marks: 10)

Evaluate the following limit

$$\lim_{y \to -2} g(y) \quad where, \; g(y) = \begin{cases} y^2 + 5 & if \; y < -2 \\ 3 - 3y & if \; y \geq -2 \end{cases}$$

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